The Role of AI in Shaping Perceived Academic Performance: Insights from Engineering Students in Sudurpaschim Province, Nepal

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ABSTRACT AND KEYWORDS

Generative AI tools such as ChatGPT are rapidly reshaping higher education globally, yet their influence on engineering students' perceived academic performance in resource-constrained, non-Western settings remains largely unexplored. This mixed-methods study investigates how engineering undergraduates in Sudurpaschim Province, Nepal perceive the impact of generative AI on their academic success, satisfaction, confidence, motivation, and study efficiency. Grounded in the Technology Acceptance Model (TAM), the study employs a pilot-tested online questionnaire combining five-point Likert scales with open-ended questions, administered to 180 third- and fourth-year engineering students across two contrasting campuses: NAST Dhangadhi (urban, stable connectivity) and Far Western University Mahendranagar (semi-urban, limited connectivity). Quantitative data will be analyzed using descriptive statistics and Cronbach's alpha reliability testing, while open-ended responses will be analyzed through Reflexive Thematic Analysis following Braun and Clarke (2006). The study addresses a critical gap: no prior study has examined these dynamics among engineering students in Sudurpaschim Province, where infrastructure constraints and lecture-centered pedagogy may moderate the perceived benefits and risks of AI adoption. Findings are expected to yield context-specific evidence to guide responsible, equitable, and pedagogically sound integration of AI in Nepalese engineering education.

Keywords: ***AI, Perceived Academic Performance, Engineering Students, Sudurpaschim Province, Nepal, ChatGPT, Study Habits, Qualitative Insights, AI in Education***

TABLE OF CONTENTS

ABSTRACT AND KEYWORDS i

TABLE OF CONTENTS ii

LIST OF TABLES iii

Chapter 1: INTRODUCTION 1

1.1 Background of the Study 1

1.2 Literature Review 2

1.3 Research Gap Analysis 3

1.4 Objectives 5

1.4.1 General Objective 5

1.4.2 Specific Objectives 5

1.5 Research Hypotheses 5

Chapter 2: METHODOLOGY 6

2.1 Description of Research Station 6

2.2 Research Design, Sample Size and Sampling Strategies 6

2.3 Study Area 7

2.4 Research Tools, Techniques and Data Collection Strategies 8

2.5 Data Management and Analysis Strategies 10

2.5.2 Analysis Strategies 10

2.5.2.1 Descriptive Quantitative Analysis (Likert-Scale Data) 10

2.5.2.2 Qualitative Analysis (Open-Ended Responses) 11

Chapter 3: EXPECTED OUTCOME AND IMPACT OF THE RESEARCH 13

Chapter 4: APPLICATION OF THE PROPOSED RESEARCH 15

Chapter 5: DELIVERABLES OF THE RESEARCH PROJECT 18

Chapter 6: PROJECT PLANNING AND MILESTONES 22

Chapter 7: BUDGET DISTRIBUTION PLAN 22

Chapter 8: RESEARCH DISSEMINATION PLAN 23

Chapter 9: RELEVANCE TO THE UN SUSTAINABLE DEVELOPMENT GOALS 25

REFERENCES 27

LIST OF TABLES

Table 6.1: Gantt Chart.....22

Table 7.1: Test Cases..... 23

Chapter 1

INTRODUCTION

1.1 Background of the Study

Generative artificial intelligence (AI) tools, such as ChatGPT and large language models, have been generating many different innovations in the field of higher education such as: personalized feedback, and explanations; help in designing a study plan; summarizing the main ideas of the content; and more. As generative AI tools offer more human and individualized support, they can be seen as a learning aid rather than another learning object. In contrast to digital learning objects that are designed to present information, generative AI tools facilitate knowledge construction. Students can use them to: - Elaborate and question their knowledge and conceptions; - Plan their studies; -Work on complex problems in a deterministic context like engineering; [1], [9].

Researchers discovered that university students who used these tools reported a number of advantages. A mixed-methods study applied to 71 students in various courses (mostly STEM, such as engineering and computer science) discovered that 83% of them said they had achieved some level of academic success using the tool, while 48% said they achieved a significant academic success and 35% achieved a slight one. Students generally reported that the tool aided them in organizing their study schedule (median score: 4.37/5), in motivating them (median score: 4.17/5) and in increasing their feelings of comfort (median score: 4.31/5). They also worked less but more efficiently and were more confident in their learning achievements. The engineering and technical disciplines were where the tool worked the best, as the tool aided students in explaining concepts and ideas and in getting sufficient feedback from their professors and teaching assistants in order to deal with the heavy workload. Likewise, students considered the tool a “supplemental” resource rather than a “replacement” and were concerned that heavy use of it would lead to students relying too heavily on technology and not doing the necessary work to think deeply and get truly involved in the course material.

The perception towards AI is highly dependent on the context. While AI is believed to enhance the personalized support for individuals in developed countries, similarly in developing countries such as Nepal, uneven infrastructure, unreliable internet, and the prevailing lecture-based pedagogy in engineering education could hinder the expected benefits while maximizing the efficiency gain for few while posing the barriers and

dependency for others. For instance, in Sudurpaschim Province of Nepal, the engineering students of National Academy of Science and Technology (Dhangadhi) and Far Western University (Mahendranagar) rely heavily on the AI for the programming tasks, preparation for the exams, solving the numerical problems, and the documentation of the projects due to the limited resources available.

Although there is an increasing number of studies suggesting that people perceive that AI can motivate themselves and organize their daily activities all over the world [1], qualitative studies are still scarce in low-resource, non- Western settings. Similarly, there is no such study done in the context of engineering students in Nepal to investigate the perceptions towards AI and how it can affect their academic success, learning satisfaction, and confidence. This is a serious concern as engineering education primarily focuses on the acquisition of knowledge and inculcating reasoning skills. Student's subjective experiences in this regard need to be investigated in depth so that their lived experiences could be taken into consideration while integrating AI in engineering education in Nepal. A mixed study carried out among the engineering students of Sudurpaschim Province of Nepal can contribute significantly towards addressing this concern in the context.

1.2 Literature Review

Artificial Intelligence in Education (AIEd) has evolved from rule-based intelligent tutoring systems (ITS) that modeled learner cognition for adaptive, real-time feedback and personalized tutoring [5], to generative large language models (LLMs) enabling dynamic content creation, interactive explanations, and collaborative knowledge support [6]. Early frameworks highlighted scalability for engagement and efficiency but raised concerns about dependency and limited independent reasoning [5], [6].

The emergence of AI tools like ChatGPT has expanded research on their role in higher education, particularly in shaping perceived academic performance student'ssubjective sense of success, motivation, efficiency, confidence, satisfaction, and overall learning progress. Studies identify common usage types: information seeking/explanations for conceptual clarification, content generation/summarization, language refinement/academic writing support, coding/programming assistance, iterative prompt refinement, planning/organization, meta-cognitive engagement (e.g., reflection on learning), and conversational repair [1]. In engineering and computing contexts, tools reduce syntactic barriers in coding/debugging and support technical problem-solving, leading to perceived efficiency gains for novices [9].

Perceived impacts vary by usage type and context. Structured AI-supported environments enhance student engagement, motivation, and perceived outcomes when integrated pedagogically [2]. In engineering education, AI-assisted instruction positively influences learning behaviors, competence development, and self-perceived achievement through personalized pathways [various engineering-focused findings, aligned with [2]]. However, substitutional use (minimal cognitive effort) may negatively affect higher-order thinking, critical reasoning, and deep synthesis [3]. Students report shifts in academic behavior, including evolving perceptions of originality, authorship, and integrity, with calls for AI literacy and process-oriented assessments to mitigate misuse [4], [10]. Broader discussions stress ethical governance and redesigned evaluations to balance benefits with risks like diminished originality [10].

In discipline-specific and regional contexts, AI supports engineering by facilitating debugging, conceptual mastery, and productivity, yet excessive reliance risks superficial understanding and reduced algorithmic reasoning [9]. Digital technologies can democratize support but risk widening inequalities due to infrastructure and literacy gaps [7]. South Asian higher education faces uneven ICT integration, faculty preparedness issues, and access barriers that moderate adoption and perceptions [8].

Despite evidence linking usage types (e.g., clarification/feedback for motivation, coding aid for efficiency) to positive perceived performance in STEM/engineering [1], [2], [9], qualitative insights into subjective experiences remain limited in low-resource, non-Western settings. Most research draws from high-income contexts, with scarce attention to how AI shapes perceived success amid Nepal's lecture-centered engineering programs, unreliable connectivity, and resource constraints. No studies examine these dynamics among engineering students in Sudurpaschim Province, where local factors (e.g., campus variations, digital divides) may influence usage types and perceptions of satisfaction/confidence. A qualitative exploration of student's lived insights is essential to contextualize global findings and inform responsible, equitable integration in Nepalese engineering education.

1.3 Research Gap Analysis

Even though the global literature on AI in higher education is burgeoning rapidly, there is still a significant underexplored area concerning how AI may shape student's perceived academic performance as well as subjective sense of success, satisfaction, confidence,

motivation, and efficiency among engineering students in a resource-constrained context like Nepal.

a. There is limited local evidence on the perceptions and experiences of the use of AI tools as most studies come from well-resourced, high-income countries [1]. It is not clear how the use of AI tools in the engineering education of Nepal impacts the perceived learning processes and study habits of students, their confidence in problem-solving, and their perceived academic gain.

b. For specific engineering education gaps, student's use of AI in tasks that are specific to a discipline (for example, for a conceptual clarification, for a coding activity, for help with system design, for numerical problem-solving, for technical report writing and for exam preparation) not known [9]. How do students use AI? What do we know about student's use of AI? Patterns in use- how do those translate into changes in subjective perceptions of performance and depth of learning?

c. Influence of local contextual factors: Unequal access to the internet, low availability of devices, financial setback, difference in infrastructure (for example, comparatively urban Dhangadhi and semi-urban Mahendranagar campuses) may moderate the use of AI tools and their benefits or widen the gap in self-assessed success [7].

d. Ethical and perceptual considerations. The Nepali institutions do not have formal guidelines on the use of AI. There is little evidence regarding engineering student's perception of academic integrity and dependency risk, ethical limits, and how this affected their feel for authentic success.

e. A connection between the AI use and the perceived performance is that there is not sufficient qualitative data exist on how the nature and extent of AI use relate to the student's self-perceived academic performance, motivation and satisfaction avoiding the use of biased objective measures like GPA.

These gaps are particularly pronounced in Sudurpaschim Province, where no region-specific studies have explored engineering student'slived insights into AI's subjective role. Addressing them through qualitative exploration of perceptions and experiences is essential to provide context-specific evidence for responsible AI integration in Nepalese engineering education [3].

1.4 Objectives

1.4.1 General Objective

The goal of the study is to determine the effect of AI-based learning tools on perceived academic performance subjective success, satisfaction, confidence, motivation, and efficiency of engineering undergraduates in Sudurpaschim Province of Nepal.

1.4.2 Specific Objectives

i. Identify the extent of AI tool usage among engineering students for different purposes, including concept grasp, problem-solving, programming, and exam preparation.

ii. Understand the changes in study habits resulting from AI, especially in terms of student's independent problem-solving, reading of textbooks, and referring to other sources

iii. Investigate engineering student's academic-related perceptions of AI tools, such as their confidence motivation satisfaction, and effectiveness.

1.5 Research Hypotheses

Grounded in the Technology Acceptance Model (TAM) and informed by the literature review and identified gaps in engineering education in Nepal, the following research questions guide this study:

i. H1 (AI Utilization in Academic Activities): Engineering students in Sudurpaschim Province regularly make use of generative AI tools for a variety of academic activities, including conceptual understanding, solving numerical problems, coding, assignment completion, composing laboratory reports, and examination preparation.

ii. H2(AI Utilization and Modification of Study Practices): The utilization of generative AI tools for academic activities impacts engineering student'sstudy practices in terms of a reduced dependency on conventional study materials (e.g., textbooks, lecture notes) and a higher engagement with alternative digital study materials.

iii. H3 (AI Utilization and Perceived Academic Performance): The utilization of generative AI tools for academic activities has a positive impact on engineering student'sperceived academic performance in terms of success, learning confidence, motivation, satisfaction, and efficiency in academic tasks in engineering fields.

Chapter 2

METHODOLOGY

2.1 Description of Research Station

The research will be a collaborative work of the National Academy of Science and Technology (NAST) that is located at Dhangadhi-4 Uttarbehedi Kailali, Nepal. The NAST will act as the main administrative and logistical centre for the research program. NAST will provide the necessary infrastructure, academic guidance, and institutional support for a research project involving engineering students of Sudurpaschim Province in different locations. NAST's facilities will be used for the project administration, data processing, and the provision of a secure room for keeping sensitive research materials such as consent forms and interim data files whose access will be limited to authorized research staff only. Data gathering at physically remote campuses will be carried out through protected online surveys and secure video conferencing methods to preserve confidentiality and uniformity. Ethical oversight will be provided by NAST's Research Division and Panel of Experts. The ethics committee of the authorized institutional body will grant the formal ethical approval besides the written consent from all the participating engineering institutions. All these research works will be carried out conforming to the ethical provisions laid down for the human-subject research.

2.2 Research Design, Sample Size and Sampling Strategies

The methodology of this study is to use a mixed-methods approach, which incorporates descriptive statistics from Likert scales and provides in-depth feedback from open-ended questions. The focus of this study is to explore and understand the experiences and opinions of third and fourth-year undergraduate engineering students in Sudurpaschim Province of Nepal regarding their experiences with generative AI in terms of their success, satisfaction, self-confidence, motivation, and efficiency.

The population of interest for this study are third- and fourth-year undergraduate engineering students in Sudurpaschim Province of Nepal, who generally have well-established study habits and substantial experiences with generative AI tools such as ChatGPT in their technical classes, such as programming, numerical problem-solving, and project works.

The sampling size for this study is 180 participants. This sampling size is deemed sufficient to detect descriptive patterns, ensure subscale reliability for items such as perceived performance using Cronbach's alpha of 0.70 or higher, and detect significant trends in Likert scale data, which is realistically obtainable in terms of face-to-face sampling in Nepal, keeping in view usual sampling response from universities.

Sampling Approach: A hybrid approach of convenience sampling and purposive sampling will be employed, as this approach allows us to incorporate both aspects of easy availability as well as trust-based relationships, while ensuring diversity along important dimensions:

i. The researchers themselves, as well as trusted student intermediaries, will personally disseminate the Google Form link while visiting campuses of NAST, Dhangadhi (urban area, better connectivity); Far Western University, Mahendranagar (semi-urban area, unstable connectivity). This link dissemination process could be carried out verbally, through personal mobile device methods, or even in groups in common places such as the university cafeteria, library, or even class breaks.

ii. The purposive sampling approach will incorporate diversity along important dimensions, namely, campus location (both urban and semi-urban, to study the effect of the digital divide); usage of generative AI technology frequency, categorized into frequent and occasional/rare; perceived academic performance and satisfaction level, based on a preliminary Likert scale; gender; engineering discipline, such as computer, civil, electrical engineering; and year of study.

2.3 Study Area

The research will be carried out in two engineering campuses in the Sudurpaschim Province. The first is the National Academy of Science and Technology (NAST) located in the city of Dhangadhi, which is an urban area that has relatively stable and reliable access to the internet and digital technologies, facilitating the stable and efficient use of AI technologies. The second is the Far Western University located in the city of Mahendranagar, which is a semi-urban or district center that is known to have unreliable and unstable access to the internet and digital technologies, which is reflective of the challenges that can be encountered in such settings.

By using these two sites, the research will be able to show how the socio-technical issues such as access, support systems offered by the institutions to the students, and the urban-rural divide may influence the perspectives that the students have about the role and impact of AI

on their academic performance, which is informed by the socio-cultural characteristics unique to the Sudurpaschim Province.

2.4 Research Tools, Techniques and Data Collection Strategies

The central focus of this research will utilize an online questionnaire which combines basic design elements with advanced analytical functions to measure Sudurpaschim Province engineering student's attitudes toward using generative AI tools such as ChatGPT and Google Gemini and Microsoft Copilot and other tools. The questionnaire will be developed using Google Forms or another secure free platform which allows users to access the platform from mobile devices and laptops while experiencing interrupted internet service. The program will provide complete access to both Nepali and English languages making it possible for users to choose the language they want to use thus creating an environment which accommodates all people.

We will test the system with 10 engineering students who will not be included in our main sample before we start our research. The study participants will provide their input about confusing terms which need to be fixed and about which examples should show engineering fields in Nepal through coding Python and solving structural analysis problems to complete the study. The study will assess the reliability of Likert subscales through Cronbach's alpha testing which needs to reach a score above 0.70 for essential sections such as perceived performance while all unclear or culturally inappropriate content requires removal or rephrasing.

The questionnaire will have five main sections, using a straightforward 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree, unless it's a different format):

a. **Demographics and Access (quick multiple-choice):** Year of study, engineering branch (computer, civil, etc.), gender, how reliable your internet usually is, what devices you own, and whether your campus has any clear rules or guidelines about using AI tools.

b. **How Often and Why You Use AI (frequency options):** Never to Daily; plus checklist for purposes: Things like clarifying tough concepts, getting help with coding/debugging, solving numerical problems, writing lab reports or assignments, preparing for exams, summarizing notes, or planning your study schedule.

c. **How AI Changes Your Study Habits (Likert items):** For example, "Using AI cuts down the time I spend solving problems on my own," "I don't feel the need to read

textbooks or lecture notes as much anymore," or "I turn to YouTube or other online resources more because of AI."

d. **How You Feel About Your Academic Performance (the core Likert subscale):** Statements like:

   i. "Using generative AI will boost my confidence in grasping engineering concepts."

   ii. "These tools make me feel more motivated and satisfied with how my studies are going."

   iii. "Overall, I feel more successful in my courses thanks to AI."

   iv. "AI helps me get engineering tasks done more efficiently."

e. **Ethical and Critical Thinking Side (Likert):** E.g., "Relying too much on AI weakens my critical thinking," or "Using AI for assignments sometimes makes me worry about academic honesty or misconduct."

We will also include a few open-ended questions to let students share in their own words:

i. Can you describe one specific moment when AI really helped (or actually held back) how successful you felt in learning?

ii. What worries you most about becoming too dependent on these tools, or about the ethics of using them?

How we will actually collect the data (planned for real-world feasibility in Sudurpaschim):

i. We will share the secure Google Form link directly and in person by visiting the campuses ourselves or through trusted friends/peers. This could mean showing a QR code in the canteen, library, or during class breaks, or quickly sending it via personal chat to students who seem interested.

ii. Informed consent will be built right into the form everyone will have to click "Yes, I agree" before starting, and it will be crystal clear that participation is 100% voluntary.

iii. Responses will stay completely anonymous (no names, IDs, or emails collected).

iv. We will keep the collection window open for 2-3weeks (March–April 2026, avoiding major exams), with gentle personal reminders to those who showed initial interest.

v. If some open-ended answers feel too brief or we want deeper stories, we will reach out to about 10 diverse respondents (chosen for different usage levels or campuses)

for short, optional follow-up chats via message or quick call only if they're okay with it.

This blend will give us solid, measurable ratings (from Likert) plus real, heartfelt stories (from open-ended) exactly what we need to understand how generative AI is quietly shaping the way engineering students here feel about their learning journey, especially with Nepal's growing focus on responsible AI use and the challenges of uneven access in places like Dhangadhi and Mahendranagar.

2.5 Data Management and Analysis Strategies

2.5.1 Data Management

All the data collected will be handled in accordance with the set guidelines and principles that govern the conduct of research and the handling of data with respect to integrity, confidentiality, and safe storage. The research will be conducted by gathering questionnaire responses from engineering students about their perspectives on generative AI in the context of academic learning.

To ensure the anonymity of the respondents, the responses to the questionnaire will be generated using Google Forms, which has the capability to generate unique identification numbers for each response received. No personal information will be requested from the respondents, such as names, email addresses, and student ID numbers.

To ensure the accuracy and reliability of the data, the data set will be randomly sampled by about 7% and checked to ascertain that the responses have been successfully transferred from the Google Forms platform to the analysis files. All the steps will be documented in the research audit trail to ensure transparency and accuracy.

Following the guidelines set by most research scholars and academics, the data set will be retained in the system for five years after the completion of the research and then permanently deleted using digital deletion methods.

2.5.2 Analysis Strategies

This study will employ a mixed-methods analytical approach, combining descriptive quantitative analysis of Likert-scale responses with qualitative thematic analysis of open-ended responses. This approach enables the research to capture both measurable trends and deeper insights into students' perceptions of generative AI in their academic practices.

2.5.2.1 Descriptive Quantitative Analysis (Likert-Scale Data)

Responses to the five-point Likert scale items (1 = Strongly Disagree to 5 = Strongly Agree) will be analyzed using descriptive statistical techniques. These analyses will summarize patterns in students' perceptions of generative AI use in engineering education.

The quantitative analysis will include:

i. Mean and standard deviation calculations for individual survey items and grouped subscales (e.g., perceived academic performance, study practices, and ethical concerns).

ii. Frequency and percentage distributions to illustrate response patterns across agreement levels.

iii. Internal reliability analysis using Cronbach's Alpha to assess the consistency of multi-item constructs. A reliability coefficient greater than 0.70 will be considered acceptable.

Data visualization techniques such as bar charts, stacked charts, and comparative graphs will be used to present response trends clearly. The statistical analysis will be conducted using software tools such as Microsoft Excel, SPSS, or R.

Since the study is exploratory in nature, the research will focus on descriptive interpretation of patterns rather than inferential statistical testing.

2.5.2.2 Qualitative Analysis (Open-Ended Responses)

Open-ended responses will be analyzed using Reflexive Thematic Analysis, following the six-phase framework proposed by Braun and Clarke (2006).

The analysis process will include:

i. Familiarisation with the data through repeated reading of responses.

ii. Generation of initial codes to identify meaningful patterns in the dataset.

iii. Searching for themes by grouping related codes.

iv. Reviewing themes to confirm their relevance and coherence.

v. Defining and naming themes to clearly describe the central meaning of each theme.

vi. Producing the analytical report, supported by representative participant quotations.

The coding process will follow an inductive approach, allowing themes to emerge directly from the data while remaining aligned with the study's research objectives.

Throughout the analysis process, the researchers will maintain reflexive notes to acknowledge their perspectives and ensure transparency in interpretation.

INTEGRATED SYNTHESIS OF QUANTITATIVE AND QUALITATIVE FINDINGS

To achieve the integrated synthesis of the findings, a convergent mixed-methods approach, as per Creswell & Plano Clark, shall be employed, where the findings of both the qualitative and quantitative aspects of the study are juxtaposed to arrive at a more comprehensive understanding of the experiences of engineering students with regard to the usage of generative AI tools.

To achieve this, the following methods shall be employed:

a. Display of tables of findings, juxtaposing the statistical findings with quotations.

b. A narrative approach, where the findings are compared, explaining the qualitative findings.

c. Triangulation of findings, where the findings are compared to establish credibility.

Such an approach shall allow a nuanced understanding of the experiences of engineering students with regard to the usage of generative AI tools, as they perceive their academic performance, study habits, and ethical considerations in the context of Nepalese education systems.

Chapter 3

EXPECTED OUTCOME AND IMPACT OF THE RESEARCH

The proposed study aims to produce systematic and contextually informed findings on the impact of AI tools (e.g., ChatGPT and other associated LLMs) on engineering student'sperceptions of their academic performance in Sudurpaschim Province, Nepal. By placing emphasis on student'ssubjective self-perceptions of their success, satisfaction, conceptual understanding, motivation to learn, efficiency in completing tasks, and overall feelings of achievement, which are to be measured using Likert scales, it is hoped to produce findings that reveal the impact of AI tools in a resource-constrained engineering education setting in a developing country.

The anticipated outcomes include:

i. **Detailed mapping of AI usage patterns and perceived influences**: The proposed study aims a comprehensive map of how engineering students in Sudurpaschim Province use generative AI tools across different academic tasks including conceptual clarification, coding, numerical problem-solving, report writing, and exam preparation and document how these usage patterns are perceived to influence academic outcomes. By capturing both frequency and purpose of use across two contrasting campuses (NAST Dhangadhi and Far Western University Mahendranagar), the study will reveal which types of AI engagement are most strongly linked to positive perceptions of learning performance, and which patterns raise concerns about dependency or surface-level engagement.

ii. **Rich insights into perceived academic performance**: In this context, this study aims to explore, through self-assessment tools in the form of Likert scales accompanied by open-ended questions for reflection, the ways in which students perceive AI as either enabling or limiting their self-perceptions of success in their studies. The expected results of this study are expected to show positive self-assessment in terms of increased confidence, motivation, satisfaction, and efficiency in handling engineering tasks, but also negative self-assessment in terms of decreased depth of involvement, critical thinking, and independent thought when using AI to the point of routine or automatic use.

iii. **Exploration of perceived ethical dimensions and integrity attitudes:** The investigation will shed light on the students' subjective understanding of the risks of dependency, challenges to originality, boundaries of academic integrity, and the ethical dimensions of

AI assistance. It will also help to determine the students' perceived needs for clearer policies and support for AI literacy and assistance.

iv. **Contextualized understanding of regional variations**: By examining the perceptions in both institutions, namely, NAST Dhangadhi, where stable connectivity is relatively guaranteed, and Far Western University Mahendranagar, where unstable connectivity was a common occurrence, the study will reveal the moderating effects of socio-technical factors, such as differential access to the Internet, device possession, financial issues, and lecture-oriented instructional methods, on the perceived benefits and drawbacks of AI by students in Sudurpaschim Province.

Overall, the study contributes to the relatively limited body of empirical literature on artificial intelligence in higher education in developing countries, with a special emphasis on engineering fields in under-resourced countries such as Nepal. The results are expected to provide valuable guidance for educators, institutions, and policymakers to promote socially responsible, equitable, and culturally sensitive use of AI tools by maximizing positive aspects such as motivation, self-confidence, and efficiency, and minimizing negative aspects such as dependency, superficiality, and ambiguity, in ways that preserve academic quality and integrity in Nepalese engineering education. By focusing on the experiences and perceptions of students in geographically and infrastructurally challenging environments, this localized approach also contributes to global debates on AI in education.

Chapter 4

APPLICATION OF THE PROPOSED RESEARCH

The results of this study are expected to be beneficial in providing guidance for engineering education in Nepal, with special emphasis on Sudurpaschim Province in the country. The study attempts to produce localized knowledge about the ways in which engineering students perceive the influence of generative AI applications like ChatGPT, Gemini, and Copilot on their success, motivation, self-confidence, efficiency, and satisfaction. By doing so, the study attempts to produce knowledge that can lead to positive changes at different levels, including students, educators, and the academic system as a whole.

From the perspective of students, the purpose of the present study is to empower engineering students through the exploration of their own utilization of artificial intelligence and their own perspectives on artificial intelligence. By understanding the ways in which these tools can be perceived as contributing to the development of confidence, motivation, satisfaction, and efficiency in the completion of day-to-day activities such as concept explanation, coding, problem-solving, report writing, and test preparation, the students can develop balanced practices in the utilization of artificial intelligence as a tool rather than a dependence, while maintaining originality and ethical decision-making during the learning process.

Instructors in engineering disciplines will have a more nuanced understanding of student's actual experience with generative AI tools. Understanding the benefits and drawbacks for students, as perceived by them such as motivation and time efficiency for the students, but superficial critical thinking and shallow work as the consequence of student's dependence on AI can help instructors improve their teaching methodologies.For example, instructors can utilize AI as a supplementary tool for student clarification and planning purposes and move towards more process-oriented evaluations like journals, oral exams, and AI-free problem sets to encourage students to think deeply and genuinely engage with the material. This could encourage students to consider artificial intelligence (AI) as a legitimate learning partner instead of an educational shortcut. The findings of this study can be beneficial for educational institutions in Sudurpaschim Province, such as NAST Dhangadhi, Far Western University Mahendranagar, and other higher educational institutions in Nepal, in formulating normative guidelines and mechanisms related to AI in the following ways:

i. Institutions can develop guidelines for the use of AI in academic studies and examinations;

ii. Institutions can develop AI literacy for students, particularly for those in engineering disciplines;

iii. Institutions can address infrastructural barriers through targeted interventions; and

iv. Institutions can encourage the proper and ethical use of AI in the educational process, thereby increasing student satisfaction and engagement.

v. The findings of this study are consistent with the National AI Policy of Nepal, which was approved in August 2025 and promotes the development of ethical guidelines, literacy, and strategies for addressing digital inequalities in education.

Such efforts are in line with the National AI Policy of Nepal, which has been ratified in August 2025 and promotes the development of ethical policies, literacy programs, and reduction of digital inequality in education.

In the area of policy formulation and curriculum development, the evidence provided in this study can be generalized and contribute to the overall reform of Nepalese engineering education, especially in relation to the Pokhara University model or other relevant frameworks.In regard to artificial intelligence in environments with limited resources, this study supports the following recommendations:

i. The development and integration of the concept of AI literacy as an essential skill in the national education policies related to engineering graduates.

ii. The development and establishment of national and regional policies related to the ethical usage of AI in higher education institutions.

iii. Advancements and developments in methodologies that seek to create an optimal balance between the use and application of artificial intelligence and integrity and deep learning.

iv. Advancements and developments in strategies that seek to address issues of digital disparities, such as the enhancement and provisioning of internet facilities in institutions of higher learning and the provision of subsidized devices.

The current study seeks to make a positive contribution to this process of providing clear, evidence-based recommendations on the responsible, equitable, and informed usage of

generative AI technologies in Nepalese engineering education. By centering the lived experiences of engineering students in the Sudurpaschim Province of Nepal, within the context of the changing Nepalese AI environment, including the recently launched National AI Policy of 2025, as well as new initiatives such as Pokhara University's newly launched Bachelor of Arts in Artificial Intelligence and Data Science, this study hopes to make a practical contribution to the improvement of student outcomes, student engagement, academic standards, and the ability of Nepalese engineering students to successfully compete in an AI-saturated world without compromising critical thinking skills and academic integrity. This study hopes to make a valuable contribution to the national conversation regarding the ethics of digital transformation in Nepalese education at the university level.

Chapter 5

DELIVERABLES OF THE RESEARCH PROJECT

The anticipated research project is expected to produce a coherent body of high-quality academic outputs that satisfy the evaluation criteria and requirements of the Department of Research and Human Resource Development (HRD), National Academy of Science and Technology (NAST). At the same time, it will produce localized perception-based evidence on the impact of artificial intelligence tools on perceived academic performance of engineering students in Sudurpaschim Province, Nepal. The key deliverables are as follows:

a. Final Research Report (Comprehensive Project Report): A comprehensive research report in a format suitable for a research report by a student would be prepared in accordance with NAST guidelines and standards for funded or HRD-supported student research projects. The research report would include: introduction where background, rationale, and significance would be presented in relation to Nepalese engineering education; literature review; methodology where a Likert scale questionnaire approach would be explained in detail; results and analysis where results obtained from Likert scale questionnaire analysis would be presented along with discussion; conclusion; limitations; recommendations for future research would be included in the report. The research would be focused on student'ssubjective perceptions of success, satisfaction, confidence, motivation, and efficiency in their studies in engineering fields.

b. Project Proposal and Progress Documentation: An approved research proposal, with the revised title, objectives, methodology using the Likert scale, ethical consideration, timeline, and budget, along with periodic progress reports to be submitted to the Research and HRD Department of NAST and the supervisor/expert panel, will be maintained. This is to ensure that systematic planning, monitoring, achievement of milestones, and compliance with NAST standards are being met in an effective manner.

c. Project Presentation and Viva-Voce/Defense Materials: Professionally designed presentation slides in a format similar to PowerPoint summarizing the research should be prepared. The content for these slides should include: background and rationale, title and objectives, methodology (including Likert scale questionnaire, sampling, data collection), key findings (including Likert scale trends on perceived performance, thematic findings from open-ended questions, regional findings), conclusions, and recommendations. The materials

should be used for presenting to the NAST Research and HRD Department, supervisor, expert panel, and possibly for a viva voce or defense.

d. Data Collection Instrument: A fully developed and pilot-tested online questionnaire, available in Nepali and English languages, will be used as the main tool for data collection. The tool will include:

i. Demographic and access questions

ii. A list of questions about usage frequency and purpose

iii. Five-point Likert scales for measuring perceived changes in study habits, perceived academic performance in terms of success, satisfaction, confidence, motivation, and efficiency, and perceived critical thinking/ethics

iv. Open-ended questions to ensure depth and student voices

v. The questionnaire, pilot test report, and results showing Cronbach's alpha values greater than 0.70 for all subscales, and any subsequent changes will be documented, appended to the report, and submitted to NAST HRD as part of the deliverables.

e. Analyzed Data and Empirical Results: A cleaned and anonymized dataset of the questionnaire results is analyzed using:

i. Descriptive statistical analysis (means, standard deviations, frequencies, and percentages) for Likert-scale and subscale results.

ii. Reliability analysis: Cronbach's Alpha Coefficients of 0.70 or Higher for Subscales: Perceived Academic Performance.

iii. Visualization of results: Bar charts, stacked bar charts, and heat maps to visualize patterns and trends, such as usage frequency in relation to perceived success ratings, etc., as they apply to specific campuses.

iv. Thematic analysis and quotes from results of the open-ended questions.

v. The results are presented in tables, figures, and written form in the final report, thereby forming the empirical base for understanding the role of AI in Sudurpaschim's engineering education, as perceived by students and staff. The anonymized and cleaned dataset will be handed over to NAST HRD in an appropriate form for possible secondary use or archiving purposes.

f. Findings and Recommendations: This study offers a clearly articulated set of conclusions and recommendations that are relevant to the context, based on the synthesis of descriptive Likert trends and thematic analysis. The recommendations offered are applicable to:

i. Students: In order to encourage ethical, balanced, and self-directed consumption of artificial intelligence, which can lead to perceived success, motivation, and self-confidence.

ii. Instructors: Facilitating accommodations in instruction and evaluation through perceived benefits while protecting critical thinking skills and academic integrity.

iii. Institutions (e.g., NAST, Far Western University): Crafting recommendations for policies aimed at addressing the digital divide in SudurPaschim Province of Nepal.

iv. Policymakers and national/regional education authorities: Recognition of AI literacy as a core competency in developing frameworks for the integration of artificial intelligence in Nepalese engineering and technology education systems.

Chapter 6

PROJECT PLANNING AND MILESTONES

The proposed research project is expected to be completed within a four-month period from February 2026 to May 2026. The timeline is structured and time-bound to ensure systematic progress, regular monitoring by the NAST Research and HRD Department, and timely submission of all deliverables in accordance with NAST guidelines for student research projects.

ACTIVITIES	Feb		Mar				April				May			
	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
Proposal approval														
Literature review														
Methodology design														
Questionnaire development														
Data collection														
Data processing														
Data analysis														
Report writing														
Revision and feedback														
Presentation slides														
Final report submission and viva-voce														

Table 6.1: Gantt Chart

Chapter 7

BUDGET DISTRIBUTION PLAN

This research, conducted in affiliation with the National Academy of Science and Technology (NAST) under the Department of Research and HRD, requires a modest budget of NPR 55,000 to support essential academic, logistical, and administrative activities for Likert Scale questionnaire-based research in Sudurpaschim Province.

S.N.	Budget Item	Description	Estimated Cost (NPR)
1	Project-related academic fees	Proposal submission, project registration, and evaluation-related fees	4,000
2	Data Collection & Connectivity	Internet and data access for literature review, online surveys (Google Forms), AI tool usage, communication, plus survey administration, participant coordination, and any travel/coordination for follow-ups	18,000
3	Academic Materials & Documentation	Books, reference materials, paid academic resources, printing of questionnaires/reports, binding, stationery, and related documentation	18,000
4	Dissemination, Presentation & Technical Support	Costs for presentation preparation (slides, printing posters if needed), seminar participation, dissemination activities, plus laptop/software usage and basic technical resources for data analysis/reporting	12,000
5	Miscellaneous expenses	Unforeseen research-related expenses	3,000
Total Estimated Budget			**55,000**

Table 7.1: Budget Distribution Plan

Source of Funding: Primary funding for this research project will be provided by the Research and Human Resource Development (HRD) Department of NAST.

Chapter 8

RESEARCH DISSEMINATION PLAN

The results of the study, which will focus on the influence of AI tools on the perceived academic performance of engineering students in the Sudurpaschim Province of Nepal, will be disseminated through various academic and institutional means. This dissemination method is intended to ensure that the localized and perception-based results of the study reach the relevant stakeholders in the academic community in Nepal, including students, instructors, university administrators, NAST HRD, policymakers, and the engineering academic community.

All the dissemination activities will be conducted ethically and will be based on the ethical guidelines of the study, NAST institutional guidelines, academic integrity guidelines, and ethical guidelines for the disclosure of AI results. Care will be taken to ensure the accurate portrayal of the subjective perceptions of the participants and the localized results of the study.

Key Dissemination Channels and Activities

a. Oral Presentations and Institutional Sharing

i. Formal presentation and defense at NAST (Dhangadhi) in front of the Research and HRD Department, the supervising committee, an expert group, and department faculty members.

ii. Departmental seminars/knowledge-sharing sessions at NAST and Far Western University (Mahendranagar), involving the students and instructors in the department of engineering and the university administration.

iii. Informal sessions for the students in the department of engineering at both the campuses to create more awareness about the effective use of AI in the context of the research findings.

b. Poster Presentations

i. Poster presentation at the research forums held at the students' research sessions at NAST, departmental poster sessions at the university and other research institutions affiliated with Pokhara University.

ii. Poster presentation at the national-level research exhibitions and/or research forums held at the Nepal Engineering Student Conference and other such events at NAST and/or at Pokhara University campuses, using simple graphics to show the trends in the Likert Scale responses on performance, emerging themes in the open-ended responses, and recommendations for the engineering education sector in Sudurpaschim.

c. Journal Publication

i. Under the guidance of the supervisors, the essential findings of the study will be crystallized into a brief manuscript suitable for submission to the Journal of NAST College (ISSN: 3102-0283, Print), the official journal of the National Academy of Science and Technology.

ii. The manuscript will highlight the study's contribution to the understanding of AI perceptions in a context of engineering in a developing country, including Likert scale patterns and open-ended findings on ethical considerations and regional variations.

iii. Depending on the findings, subsequent journal publications in various national journals in educational technology or engineering education will be explored after publication in NAST College.

Every dissemination strategy employed will ensure transparency, confidentiality of participants, cultural sensitivity, and clear explanations of the exploratory, perception-based approach of the study. Through dissemination of the study through NAST-centered channels as well as regional channels, the study hopes to achieve its relevance, credibility, and impact on the ethical incorporation of AI in Nepalese engineering education.

Chapter 9

RELEVANCE TO THE UN SUSTAINABLE DEVELOPMENT GOALS

The proposed study **"The Role of AI in Shaping Perceived Academic Performance: Insights from Engineering Students in Sudurpaschim Province, Nepal"**, strongly aligns with several of the United Nations Sustainable Development Goals (SDGs), most directly with SDG 4: Quality Education, while also contributing indirectly to SDG 9: Industry, Innovation and Infrastructure and elements of SDG 10: Reduced Inequalities.

Primary Alignment: SDG 4 seeks to "ensure inclusive and equitable quality education and promote lifelong learning opportunities for all." The proposed study contributes to this goal by providing localized data on how engineering students in Nepal perceive the impact of artificial intelligence tools, such as ChatGPT and other variants of this technology, on their academic performance, motivation, satisfaction, and efficiency.

i. **Overall contribution to SDG 4:** The study's examination of student'ssubjective perceptions through the application of Likert scale questions and observational responses will provide regional-level data that can be used to guide the judicious, inclusive, and appropriate incorporation of artificial intelligence in higher education settings with the goal of improving the quality and relevance of the educational experiences offered in engineering courses, especially in less-resource-endowed regions such as the Sudurpaschim Province.

ii. **Alignment with Target 4.3:** "By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university." The research's examination of the role that artificial intelligence can play as an adjunct to academic support services will provide insights into the perceived benefits and challenges that can be addressed through strategies that reduce disparities in the quality and accessibility of engineering education, especially in semi-urban settings such as the Far Western University Mahendranagar.

iii. **Alignment with Target 4.7:** "By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including… education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship, and appreciation of cultural diversity and of culture's contribution to sustainable development." The

study's examination of student'ssubjective responses to the application and implications of artificial intelligence will guide the development of higher education settings that are not only ethical and socially responsible but also inclusive and balanced in terms of the role that technology can and should play in the educational process.

Indirect Contribution to SDG 9 - Industry, Innovation and Infrastructure SDG 9 seeks to "build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation."

i. By providing research-based information on how artificial intelligence (AI) is viewed as a tool for enhancing motivation, confidence, and efficiency in carrying out tasks in engineering fields such as coding, problem-solving, and writing, this study indirectly contributes to developing a skilled workforce in the field of engineering.

ii. The research findings can be used in contributing to innovation, inclusive and sustainable industrialization, and infrastructure in Nepal, with a specific focus on addressing regional digital divide issues in Sudurpaschim Province where there is a lack of internet connectivity.

Overall Contribution to the 2030 Agenda: This project contributes to the United Nations Sustainable Development Agenda by supporting the provision of excellent, inclusive, ethical, and equitable higher education through the effective utilization of artificial intelligence (AI) technologies. The results of the study from the Sudurpaschim Province can be used as a resource for Nepal as a whole, providing guidance on the effective utilization of AI while maintaining academic integrity, critical thinking skills, and inclusivity. Globally, the study is a contribution to the utilization of AI for sustainable transformation in higher education while addressing the unique challenges facing students in the developing world.

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